Automatic and Accurate Image Matting

Wu-Chih Hu**, Deng-Yuan Huang, Ching-Yu Yang, Jia-Jie Jhu, Cheng-Pin Lin

National Penghu University of Science and Technology, Taiwan, ROC

Outline

- ❖Introduction
- ❖Modified Spectral Matting
- ❖Experimental results
- ❖Conclusion

Introduction

- Image matting plays an important role in image and video editing.
- Image matting belongs to image segmentation, but it is not a binary segmentation.
- Image matting takes the problem of estimating the partial opacity of each pixel in a given image.

$$I_i = \alpha_i F_i + (1 - \alpha_i) B_i$$

Introduction (cont.)

- Image matting is a highly under-constrained problem.
  - A trimap
  - A set of scribbles (brush strokes).

Introduction (cont.)

- **The unsupervised image matting (spectral matting) was first proposed by Levin et al. [7].**
- **It is a challenging task for unsupervised image matting to obtain an accurate alpha matte.**
- **In this paper, the modified spectral matting is proposed to obtain automatic and accurate alpha matte.**

[7] Levin, A., Rav-Acha, A., Lischinski, D.: Spectral Matting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 30(10), 1699--1712 (2008).

Modified spectral matting

- In spectral matting, the compositing equation is generalized by assuming that each pixel is a convex combination of *K* image layers $F^1 \sim F^K$

$$I_i = \sum_{k=1}^{K} \alpha_i^k F_i^k \qquad \sum_{k=1}^{K} \alpha_i^k = 1;\ \alpha_i^k \in [0,1]$$

- Suppose that the input image consists of *K* distinct components $C_1 \sim C_K$ such that $C_i \cap C_j = \phi$ for $i \neq j$
- Compute the eigenvectors of $N \times N$ Laplacian matrix *L* as $E = [e^1, \ldots, e^M]$, where *E* is a $N \times M$ matrix (*N* is the total number of pixels)

Modified spectral matting (cont.)

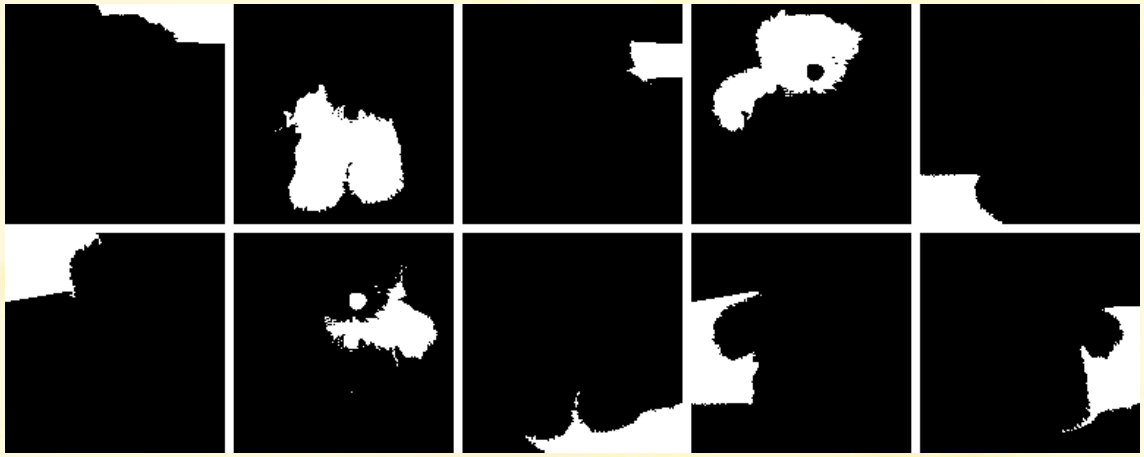
- **The distinct components are obtained using spectral segmentation with the *k*-means algorithm based on the eigenvectors of matting Laplacian.**
- **Matting Laplacian is defined as a sum of matrices $L = \sum_q A_q$, each of which contains the affinities among pixels inside a local window w_q :**

$$A_q(i,j) = \begin{cases} \delta_{ij} - \dfrac{1}{|w_q|}\left(1 + (I_i - \mu_q)^T \left(\sum_q + \dfrac{\varepsilon}{|w_q|} I_{3\times 3}\right)^{-1} (I_j - \mu_q)\right) & , (i,j) \in w_q \\ 0 & , \text{otherwise} \end{cases}$$

δ_{ij} is the Kronecker delta; μ_q is the mean color vector in the window w_q around pixel q; $\sum_q$ is a 3×3 covariance matrix in the same window; $|w_q|$ is the number of pixels in the window; and $I_{3\times 3}$ is the identity matrix.

Modified spectral matting (cont.)

- The result of distinct component detection using spectral segmentation with the *k*-means algorithm.



Modified spectral matting (cont.)

- The palette-based component classification
  - The boundary pixels (with a width of one pixel) of the input image are extracted to analysis components of RGB.
  - RGB color space of distinct component is transformed to the HSV color space.

$$H = \begin{cases} (G-B)*60/(Max-Min) & , \text{ if } R = Max \\ 180+(B-R)*60/(Max-Min) & , \text{ if } G = Max \\ 240+(R-G)*60/(Max-Min) & , \text{ if } B = Max \end{cases}$$

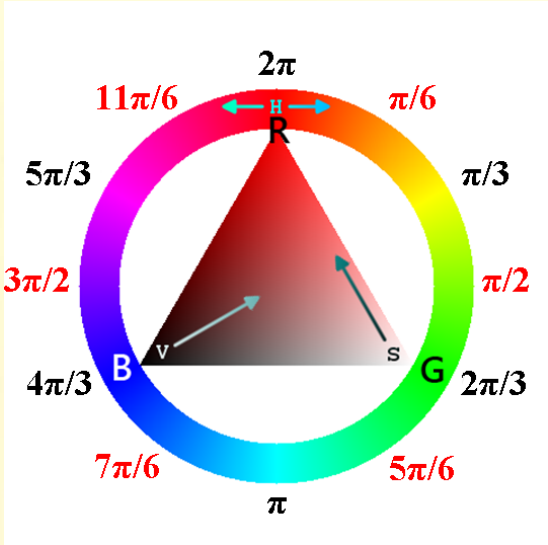
Modified spectral matting (cont.)

- Calculate the average hue $\widetilde{H}$ of each distinct component.
- The candidate components of foreground and background are obtained using

$$F: \begin{cases} \pi-\theta \le \widetilde{H} \le \pi+\theta & \text{, if pallette} = R \\ (5\pi/3)-\theta \le \widetilde{H} \le (5\pi/3)+\theta & \text{, if pallette} = G \\ (\pi/3)-\theta \le \widetilde{H} \le (\pi/3)+\theta & \text{, if pallette} = B \end{cases}$$

$$B: \begin{cases} 2\pi-\theta \le \widetilde{H} \le 2\pi+\theta & \text{, if pallette} = R \\ (2\pi/3)-\theta \le \widetilde{H} \le (2\pi/3)+\theta & \text{, if pallette} = G \\ (4\pi/3)-\theta \le \widetilde{H} \le (4\pi/3)+\theta & \text{, if pallette} = B \end{cases}$$

$\theta = \pi/6$ is set from experience

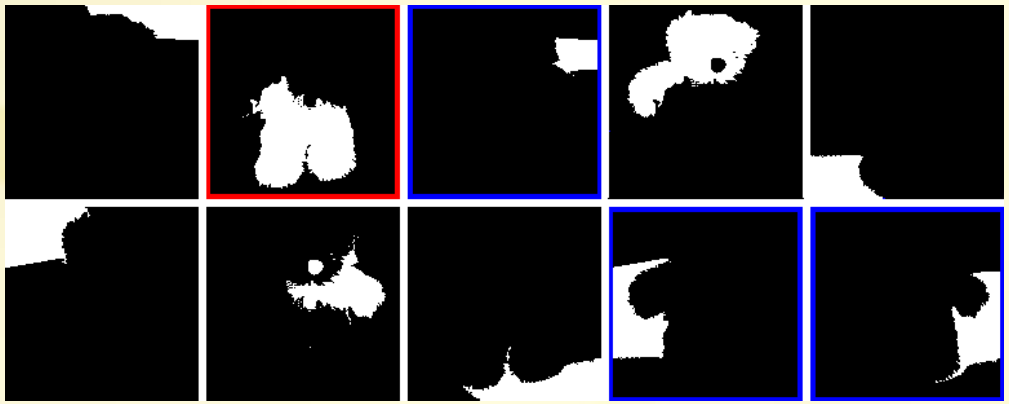


Modified spectral matting (cont.)

- These candidate components of foreground and background are checked again.
  - If the average value of the palette of the candidate component of the foreground is smaller than the average value of the palette of the input image, then it is the final component of the foreground.
  - If the average value of the palette of the candidate component of the background is bigger than the average value of the palette of the input image, then it is the final component of the background.

Modified spectral matting (cont.)

- Using the proposed palette-based component classification, the distinct components are classified as the components of foreground, background and unknown regions.



- The corresponding matting components are obtained via a linear transformation of the smallest eigenvectors of the matting Laplacian matrix.

$$\widetilde{E} = [e^1, \ldots, e^{\widetilde{M}}]$$

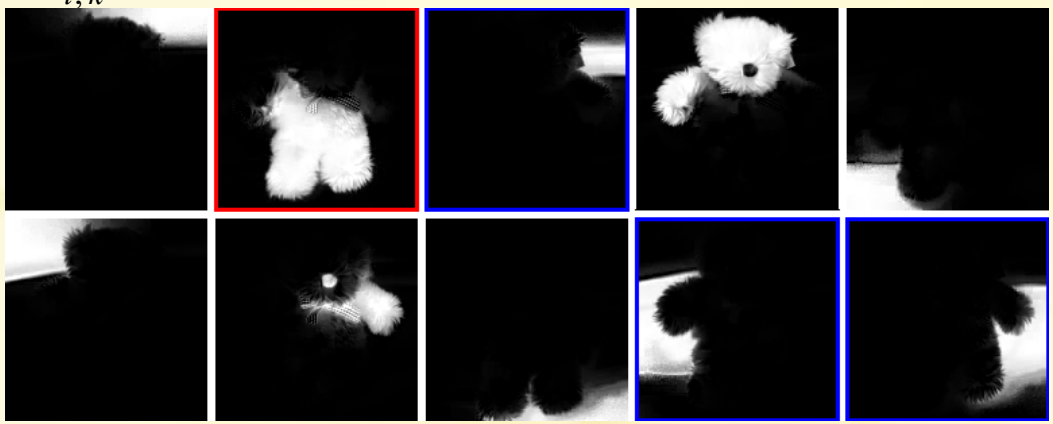
Modified spectral matting (cont.)

- **Initialize α^k by applying a *k*-means algorithm on the smallest eigenvectors, and project the indicator vectors of the resulting components $C_1 \sim C_K$ on the span of the eigenvectors using**

$$\alpha^k = \widetilde{E}\widetilde{E}^T m^{C^k} \qquad m_i^C = \begin{cases} 1 & , i \in C \\ 0 & , i \notin C \end{cases}$$

- **Compute matting components by minimizing an energy function**

$$\sum_{i,k} \left|\alpha_i^k\right|^\gamma + \left|1-\alpha_i^k\right|^\gamma, \text{ where } \alpha^k = \widetilde{E}y^k$$



Modified spectral matting (cont.)

- **Finally, only matting components of the foreground and unknown region are combined to form the complete alpha matte based on minimizing the matte cost using**

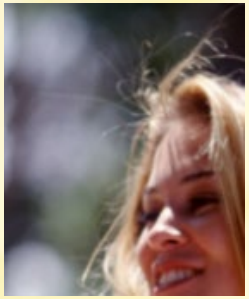
$$J(\alpha) = \alpha^T L \alpha$$

Experimental results

- The algorithms are implemented in Matlab R2008b .
- The number K of clusters using k-means algorithm is set as 10.
- The number M of the smallest eigenvectors in finding distinct components is set as 20.
- The number $\widetilde{M}$ of the smallest eigenvectors in finding matting components is set as 50.
- The ground truth of the alpha matte was obtained by manually selecting the matting components of the foreground.

Experimental results (cont.)

❖ **Image matting of a face image**



Spectral matting

Our method

Experimental results (cont.)

❖ **Image matting of a wolf image**

Spectral matting

Our method

Experimental results (cont.)

❖ **Image matting of a face image**

Spectral matting

Our method

Conclusions

- The modified spectral matting was proposed to obtain automatic and accurate image matting.
- The palette-based component classification is proposed to obtain the reliable foreground and background components.
- Experimental results show that the proposed method has better performance than the spectral matting.

Thanks for your attention !